

Multiplexed Detection of Tumor Markers and Discrimination of Cancer Cell Types by Laser Ablation-Dielectric Barrier Discharge Ionization-Mass Spectrometry

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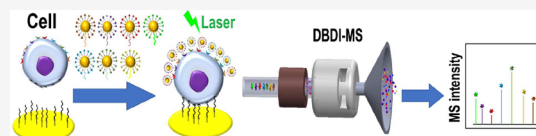
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ABSTRACT: We present an ambient mass spectrometry immunoassay platform based on specific mass tags detected by laser ablation-dielectric barrier discharge ionization-mass spectrometry (LA-DBDI-MS). It features high sensitivity, multiplexed quantitation, minimal sample consumption, and convenient operation. The organic small-molecule mass tags allow very high sensitivity and quantitative detection of multiple proteins, even of membrane-bound proteins on cell surfaces, through signal amplification (approximately 600 times). By using just 2 μ L of serum, we achieved the detection of thrombin (LOD 6.6 pM) and cancer antigen 125 (CA125) (LOD 1.4 U/mL). Seven protein biomarkers, including CA125, carcinoembryonic antigen (CEA), epithelial cell adhesion molecule (EpCAM), protein tyrosine kinase 7 (PTK7), transferrin receptor 1 (CD71), cluster of differentiation 8 alpha protein (CD8a), and cluster of differentiation 33 (CD33), were simultaneously detected in situ in four types of cancer cells within 2 h. This platform is expected to enable multiplexed protein detection in single-drop samples or at the single-cell scale, distinguish different types of cells, and has potential applications in clinical diagnosis.



INTRODUCTION

Proteins are indispensable biomolecules in living organisms that perform a diverse array of biological functions. Many proteins are required for specific activities within organisms and cells, whereas aberrant protein expression is frequently linked to diseases.¹ Consequently, the sensitive and selective detection of specific protein biomarkers is of importance for fundamental research, biomedical studies, disease diagnosis,² and biodefense applications.³ Evidence suggests that even minute quantities of a particular protein can significantly impact cellular biological functions and initiate pathophysiological processes.⁴ Furthermore, during the early stages of disease, disease-associated protein biomarkers in bodily fluids or tissues are typically expressed at very low concentrations, posing significant challenges for conventional detection methods.⁵ Hence, there is a need to enhance the detection sensitivity for specific low-abundance protein biomarkers.

Due to their convenience and exceptional sensitivity, immunoassays have emerged as a preeminent method in such applications. Specifically, enzyme-linked immunosorbent assays (ELISAs), which rely on the specific interactions between antigens and antibodies, are widely regarded as the gold standard for quantifying proteins.⁶ In addition to antibodies, aptamers (comprising short oligonucleotides of DNA or RNA) and peptides offer diverse alternatives for recognition and capture processes.⁷ However, the spectrophotometric readout of immunoassay applications is frequently hampered by issues such as band overlap,⁸ background interferences, and a narrow dynamic range, thereby limiting

their use for sensitive and especially multiplexed detection.⁹ Despite advancements in iterative staining strategies that have enhanced throughput, there remains a demand for efficient multiplexed measurements on one spot.¹⁰

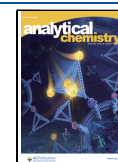
In contrast, mass spectrometry (MS) is considered an optimal readout method for immunoassays, owing to its inherent advantages of high sensitivity, a broad dynamic range,¹¹ minimal background interference, and high-throughput detection capabilities.¹² It overcomes the limitations of multiplexed detection due to its exceptional resolution. Direct MS analysis of proteins offers both sequence and quantitative information, yet the detection of large biomolecules frequently encounters challenges related to sensitivity and ionization efficiency.¹³ Consequently, indirect protein measurement through labeling strategies in MS-based immunoassays has garnered significant interest in recent years. Inductively coupled plasma (ICP) MS has been employed to detect multiple proteins tagged with various elements.^{14–16} Mass cytometry, also known as CyTOF-MS, utilizes ICP–time-of-flight instrumentation¹⁷ for detecting multiple markers simultaneously with high acquisition rate, synchronized with

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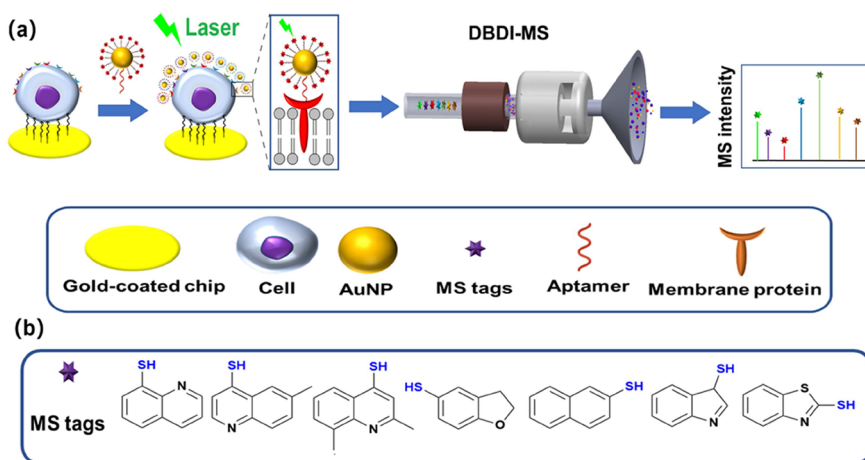


Figure 1. (a) Schematic of an AMS immunoassay platform and its workflow for protein detection on cell surfaces. (b) Chemical structures of mass tags.

the repetition rate of the ablation laser.¹⁸ This technology measures the abundance of rare earth markers tagged to antibodies, and amplification is achieved through the use of multiple metal-chelating moieties¹⁹ or sometimes metal nanoparticles.²⁰ However, the need for specialized equipment, costly elemental tags, and limited availability of labeled antibodies restrict the widespread application of CyTOF-MS.²¹ In contrast, organic molecules provide an almost unlimited spectrum of possibilities for tag design compared to elements.²² The integration of organic tags with ambient mass spectrometry (AMS) offers simple and cost-effective detection.²³ While AMS possesses numerous advantages, it faces one key challenge, namely low ion transmission efficiency.²⁴ The “active capillary plasma source”²⁵ developed by the Zenobi group dramatically improves ion transmission in AMS, and the dielectric barrier discharge ionization (DBDI) method that is part of this strategy also exhibits a good ionization efficiency for nonpolar substances.²⁶ Therefore, by incorporating suitable MS tags and applying DBDI in immunoassay analysis within AMS, the detection sensitivity can be significantly enhanced.

We have developed the use of laser desorption/ionization, followed by DBDI, which is exceptionally efficient to postionize small organic molecule tags, to conduct in situ detection of several low-abundance protein biomarkers, including cell surface proteins, as shown in Figure 1a. The laser passes through a transparent glass chamber to desorb and ionize the organic mass tags, which are then transmitted to the DBDI source for postionization. The enclosed, flow-through arrangement of the DBDI source enhances molecule/ion transmission efficiency compared to other open-space ambient ionization methods. Specific aptamers are utilized to immobilize protein biomarkers within a sandwich assay configuration, sandwiched between gold nanoparticles (AuNPs) and a gold substrate. Small molecules, as shown in Figure 1b, chosen for their high ionization efficiency and capacity to minimize nonspecific adsorption, serve as mass tags and are attached to the AuNPs to amplify the MS signal. After extensive screening, we identified several small organic molecule tags suitable for laser ablation-dielectric barrier discharge ionization mass spectrometry (LA-DBDI-MS). These tags have a heterocyclic or naphthalene skeleton, various substituents, and a thiol group to facilitate binding to

Au. By varying or increasing the number of substituents, an infinite array of small molecule mass tags can theoretically be generated. Moreover, these organic molecular tags are smaller than those previously reported,¹² allowing for more tags to be attached to AuNPs. Additionally, the AuNPs function as inorganic matrices for laser ablation, simplifying the overall experimental workflow. The mass tags attached to the AuNPs are analyzed by LA-DBDI-MS, allowing for the indirect representation and detection of the biomarkers. Another innovation is that we were able to minimize spectral background interferences, due to the sensitivity of the selected label to DBDI and the use of clean nitrogen, which cannot easily be achieved by other ambient ionization methods. The utilization of multiple mass tags, signal amplification techniques, and a high-throughput, convenient AMS method enabled the ultrasensitive and efficient detection of several protein biomarkers within single droplets of bodily fluids or on a limited number of cells in situ. Similarly, different kinds of cells can be distinguished by their cell surface antigens that lead to specific signals in the mass spectrum.

EXPERIMENTAL SECTION

Preparation of Mass-Tagged AuNPs (mAuNPs) Conjugated to Aptamers. AuNPs with an average diameter of 15 nm were prepared via citrate reduction of HAuCl₄ as described previously.²⁷ The sizes of these AuNPs were confirmed using transmission electron microscopy (TEM), and their elemental composition was verified through Energy Dispersive X-ray Spectroscopy (EDS). Surface modification of AuNPs with aptamers and mass tags was achieved using a freeze–thaw method,²⁸ following extensive research on suitable aptamers and mass tags, as shown in Figure S2. The optimized functionalization protocol for AuNPs involved several steps. Initially, thiol-modified aptamers were activated for 1 h using 10 mM freshly prepared tris(2-carboxyethyl) phosphine (TCEP). Subsequently, 20 μ L of 10 μ mol/L aptamers specific for thrombin (tApt 1), CA125 (cApt), CEA (ceApt), EpCAM (epApt), PTK7(pApt), CD71(cdApt) CD8a (cd8Apt) or CD33 (cd33Apt) were added to 1 mL of 8.2 nmol/L AuNPs with a diameter of 15 nm. The mixture was frozen at -20 $^{\circ}$ C in the dark for 6 h, thawed to room temperature for 10 min, and subjected to two complete freeze–thaw cycles. Following this, 50 μ L of 1 mmol/L mass tags

(preactivated overnight with TCEP) were added. After the addition of mass tags, the mixture was gently agitated overnight to obtain AuNPs modified with both mass tags and aptamers. Subsequently, 1 mmol/L MUDOL was used to further block free sites on the AuNPs for 2 h, aiming to reduce nonspecific adsorption. These modified nanoparticles were stored in the presence of excess aptamers at 4 °C. Prior to use, the AuNPs were centrifuged at 8000 rpm for 15 min to remove excess reagents. A salt-washing step using binding buffer was then performed to eliminate all unmodified AuNPs, followed by redispersion of the AuNPs in the binding buffer. The supernatant of the last washing was detected by MS to ensure the sufficient washing of unbound mass tags.

Chip Functionalization. Gold-coated silicon wafers were cleaned with acetone and ethanol in an ultrasonic bath for 20 min each, followed by drying in the air. To enable thrombin or cell recognition, specific aptamers were attached to the gold layer. Prior to application, 2 μ L of 50 μ mol/L tApt 2 (specific for thrombin) or mApt (specific for MUC1) were mixed with 10 μ L of 4 mmol/L TCEP for activation and then deposited onto the gold layer. The wafers were incubated at 4 °C for 16 h, washed twice with tris-HCl buffer, and subsequently blocked in tris-HCl buffer containing 1 mmol/L MCH for 2 h to minimize nonspecific adsorption. The prepared wafers were ready for immediate use or could be stored at 4 °C for up to 1 week. For the recognition of CA125 protein, the gold substrates were modified with antibodies using the common EDC/NHS coupling method.²⁹ Briefly, the gold-coated wafers were immersed in 10 mmol/L MUA overnight to functionalize the gold surface with carboxylic groups. Subsequently, the wafers were washed with water to remove unbound MUA, and the carboxylic groups were activated with EDC/NHS (0.05 g EDC and 0.01 g NHS dissolved in 4 mL H₂O) for 20 min, followed by another water wash. After depositing 2 μ L of 1 μ mol/L anti-CA125 antibody onto the layer, the wafers were incubated at 4 °C overnight. Subsequently, 10 mmol/L ethanolamine was applied to the wafers to block the remaining carboxylic groups. Additionally, the gold wafers were further passivated in 1 mmol/L MCH. The antibody-modified gold wafers were then immediately used for the desired applications.

Sample Preparation, Target Protein Recognition, and Detection. First, 2 μ L of the sample containing thrombin in PBS/serum and CA125 in PBS/serum were dispensed onto an aptamer/antibody-modified chip and incubated at 37 °C for 60 min to achieve target protein recognition. After washing with buffer, the chip carrying captured target proteins was immersed in a 60 μ L suspension of AuNPs/Apts/Tags and incubated again at 37 °C for 60 min to complete the binding with mass tag probes. Subsequently, the chip was washed twice with tris-HCl buffer and water, respectively. After drying, the sample chip was placed in the sample chamber and analyzed using the LA-DBDI-MS apparatus for mass spectrometric detection.

For the detection of cell surface proteins, a 40 min incubation time was adopted to ensure efficient protein recognition and minimize nonspecific background signals from the cytoplasm. When detecting seven targets simultaneously, seven types of AuNPs probes were mixed in a 1:1:1:1:1:1:1 ratio, and the chip was immersed in 280 μ L of the mixed probes. After incubation with AuNPs probes at 37 °C for 40 min, the chip was washed three times with PBS buffer before imaging. To assess nonspecific binding in the sample, a negative control (using only buffer/serum) was prepared in parallel. For direct detection of target proteins,

only 2 μ L of thrombin solution was dispensed onto the chip, and analyzed using the same parameters on the MS platform.

RESULTS AND DISCUSSION

Characterization of AuNPs and Quantification of Tags. AuNPs with an average diameter of 15 nm were prepared via citrate reduction of HAuCl₄ as described previously.²⁷ The sizes of these AuNPs were confirmed using transmission electron microscopy (TEM), and their elemental composition was verified through Energy Dispersive X-ray Spectroscopy (EDS), as shown in Figure S1. Surface modification of AuNPs with aptamers and mass tags was achieved using a freeze–thaw method,²⁸ which resulted in a greater number of aptamers per gold nanoparticle. The possible mechanism is that at room temperature, negatively charged cit-AuNPs and DNA repel each other. As the temperature drops, ice crystals start to gradually form, causing AuNPs, DNA, and salt ions (Na⁺ and citrate ions) to be squeezed out into “micropockets”. This upconcentration in the micropockets significantly accelerates the reaction rate. Subsequently, the AuNPs were characterized by TEM and EDS. Figure S1a,c shows data from AuNPs, and (b,d) from aptamer-labeled AuNPs. The average particle size of the gold nanoparticles was determined to be 15 nm. The successful attachment of aptamers to the gold nanoparticles is evident from changes in the sulfur S elemental content observed in the EDS analysis data presented in Table S1. UV-vis data of modified AuNPs are shown in Figure S2.

The detection system consists of a laser, a DBDI postionization source, and a mass spectrometer, as shown in Figure 2. A home-built DBDI source in the “active sampling

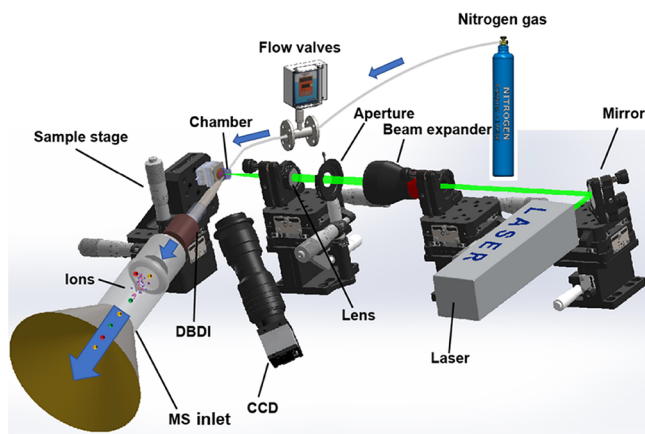


Figure 2. Schematic diagram of the detection system.

capillary” configuration was used, as described in detail in an earlier publication.³⁰ Compared with the conventional open geometry, this in-line geometry results in a very high transmission efficiency.³¹ The sample was placed inside a cylindrical quartz glass chamber (Figure S1) with an inner diameter of 1 cm to facilitate efficient transmission of the sample after laser ablation. At the same time, to avoid interferences from plasticizer molecules in the laboratory air, the carrier gas used was clean N₂, and its flow rate was controlled by a mass flow controller.

The mass tags play a critical role here as mass spectrometry reporters and for signal amplification for ultrasensitive and multiplexed protein quantification. Molecules with high

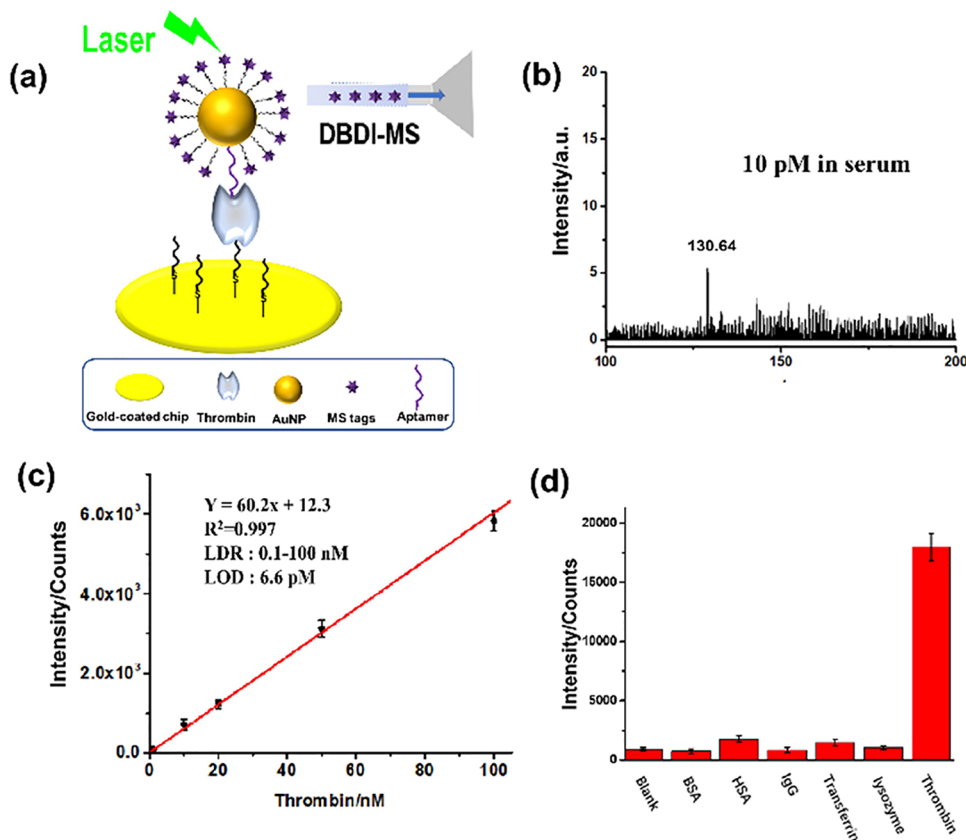


Figure 3. (a) Schematic of detection of thrombin, (b) mass spectrum from mass tags corresponding to 10 pM thrombin in serum, (c) calibration curve of thrombin in serum (0.1–100 nM), and (d) specificity test for thrombin detection. Negative results from the blank (only PBS buffer), bovine serum albumin (BSA), human serum albumin (HSA), immunoglobulin G (IgG), lysozyme, and serum without thrombin spiked show the low nonspecific binding and good selectivity of the dual-aptamer-based recognition.

ionization efficiency and good mass spectrometric response are ideal candidates. Seven small molecular mass tags that are efficiently ionized by DBDI were selected, each in the mass range from 100 to 200 Da and with minimal structural differences between them. And the molecular weights of these tags are closely spaced, allowing for optimal utilization of the mass spectrometry detection window. The detection of these mass tags involves two sequential steps. First, the tags are desorbed from the surface of the AuNPs upon laser irradiation. Subsequently, the tags enter the DBDI source, where they are ionized before being detected by the mass spectrometer, as shown in Figure S3. An intriguing phenomenon was observed: as shown in Figure S4, the mass spectral signals of the mass tags conjugated to the gold nanoparticles exhibit a loss of one S atom. Cleavage of the C–S bonds probably occurs during the laser ablation process. Additionally, we investigated the source of the extra hydrogen atoms. We conducted experiments by substituting H_2O by D_2O , as shown in Figure S5, but found no shift in the mass spectral signals, suggesting that the protons likely originate from the citrate ions present on the surface of the gold nanoparticles. To investigate the number of mass tag molecules bound to each AuNP, we displaced the mass tags on the nanoparticles with mercaptoethanol. The displaced mass tags were then quantified using liquid chromatography (LC), as shown in Figure S6.

Due to differences in the mass spectrometric response of each mass tag and variations in the number of tags attached to each nanoparticle, we calculated the relative response factor

(RRF) for each tag. The relative protein abundance can then be determined using the following formula:

$$\text{Protein abundance} = \frac{\text{Mass spectrometry signal intensity}}{\text{RRF}}$$

This approach allows for the accurate quantification of the relative amounts of different proteins based on the MS signals of the corresponding mass tag, as shown in Table S2.

Detection of Thrombin. To demonstrate the feasibility of the mass tag-based MS signal amplification method for detection of target proteins, we applied the method to thrombin, a serine protease crucial for hemostasis and blood coagulation by selectively cleaving Arg–Gly bonds in fibrinogen to generate fibrin and activate platelets.³² Aberrant overexpression or activity imbalance of thrombin is implicated in various diseases, including thrombosis, inflammation, atherosclerosis, and hemophilia. Consequently, sensitive detection of thrombin should be useful for disease diagnosis and drug discovery endeavors. By depositing a thrombin-containing solution on the aptamer modified gold substrates, thrombin can be selectively captured and then capture a thrombin-specific aptamer modified mass-Tagged AuNPs (mAuNPs) to form a sandwich structure as shown in Figure 3. The gold substrates can be analyzed directly by LA-DBDI-MS to obtain the results. As shown in Figure 3b. Even at concentrations as low as 10 pM in the serum, we could still obtain the signal at m/z 130.64 for the tag molecules that serve as reporters for thrombin. In contrast, when there was no thrombin in solution, no signal at m/z 130.64 could be observed, which

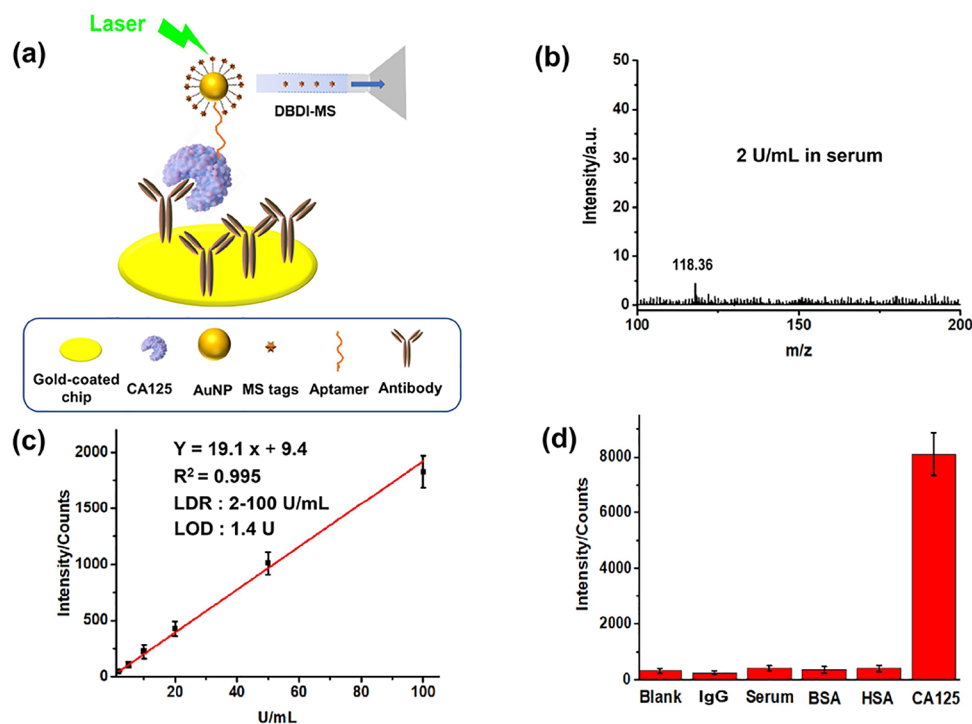


Figure 4. (a) Schematic of CA125 detection. (b) Mass spectrum of mass tags representative of 2 U/mL CA125 in serum. (c) Calibration curve for CA125 in serum (2–100 U/mL). (d) Specificity test for CA125 detection. Negative experiments with blank (only PBS buffer), immunoglobulin G (IgG), serum, bovine serum albumin (BSA), human serum albumin (HSA), and serum without CA125 spiked showed the low nonspecific binding and good selectivity.

demonstrates that the observation of tag molecules at m/z 130.64 correlated with the presence of thrombin. Moreover, an excellent linear correlation between the MS signal intensity and thrombin concentration was observed, as shown in Figure 3c. The linear detection range (LDR) was 0.1–100 nM, the R^2 was 0.997, the RSD was 9.4% ($n = 6$) and the limit of detection (LOD) was as low as 6.6 pM.

We further tested the specificity of the signal amplification assay by adding other proteins into the solution, such as bovine serum albumin (BSA), human serum albumin (HSA), immunoglobulin G (IgG) and lysozyme (instead of thrombin), using the same experimental procedures as mentioned above, as shown in Figure 3d. Negative controls with blank (only PBS buffer) and serum without thrombin spiked showed a very low degree of nonspecific binding and good selectivity of the dual-aptamer based recognition, indicating the potential value of the method as a quantitative protein assay to determine the approximate concentration of protein biomarkers.

Detection of CA125. Cancer antigen 125 (CA125) is a high molecular weight biomarker used in the diagnosis of ovarian cancer approved by the U.S. Food and Drug Administration (FDA). The mucinous glycoprotein CA125 is present on the surface of ovarian cancer cells.³³ In serum samples of patients with ovarian cancer, CA125 protein rises above 35 U/mL. Clinical assays such as enzyme, immunoradiometric, and fluorescence have been used to determine CA125 antigen.³⁴ These methods are costly, time-consuming, and require specialized personnel. For these reasons, sensitive, practical, and economical determination of CA125 antigen, critical for early ovarian cancer diagnosis, is essential.³⁵ Antibodies were immobilized on chips for recognition, confirming that this assay can be flexibly extended to other types of specific recognition as shown in Figure 4a. Even at

concentrations as low as 2 U/mL in the serum (far below the minimum value of 35 U/mL found in the serum of ovarian cancer patients), we could still obtain the signal at m/z 118.36 for the tag molecules representative of CA125, as shown in Figure 4b. Moreover, a clear correlation between MS signal and CA125 concentration was observed, as shown in Figure 4c, with a LDR of 2–100 U/mL. The R^2 was 0.995, the RSD was 9.8% ($n = 6$) and the LOD was as low as 1.4 U/mL, which fully meets the clinical diagnostic requirements of ovarian cancer patients. This highly sensitive detection method provides a powerful tool for the early diagnosis of ovarian cancer.

Detection of Cancer Cells. Since most biomarkers are found on cell surfaces,³⁶ the MS tags were evaluated for their potential in detecting multiple proteins in situ on living cells. Breast cancer cells (MCF-7), ovarian cancer cells (OVCAR-3), hepatocellular carcinoma cells (HepG2), and Jurkat cells (Jurkat) were selected as model cell lines. When 2 μ L cell suspension solutions were dispensed onto the chips, these cells were captured by mucin 1 protein aptamers.³⁷ Subsequently, the chips were incubated with AuNPs probes for the in situ recognition of target proteins. Five important cancer biomarkers localized on the cell surfaces were analyzed simultaneously in situ, such as Cancer Antigen 125 (CA125), carcinoembryonic antigen (CEA), epithelial cell adhesion molecule (EpCAM), Protein tyrosine kinase 7 (PTK7), transferrin receptor 1 (CD71), cluster of differentiation 8 alpha protein (CD8a), and cluster of differentiation 33 (CD33). These biomarkers were interrogated using seven different labeled AuNPs, which were subsequently dissociated and analyzed concurrently by LA-DBDI-MS. All MS signals were derived from the organic small molecular mass tags corresponding to the protein. Notably, the spectral signatures

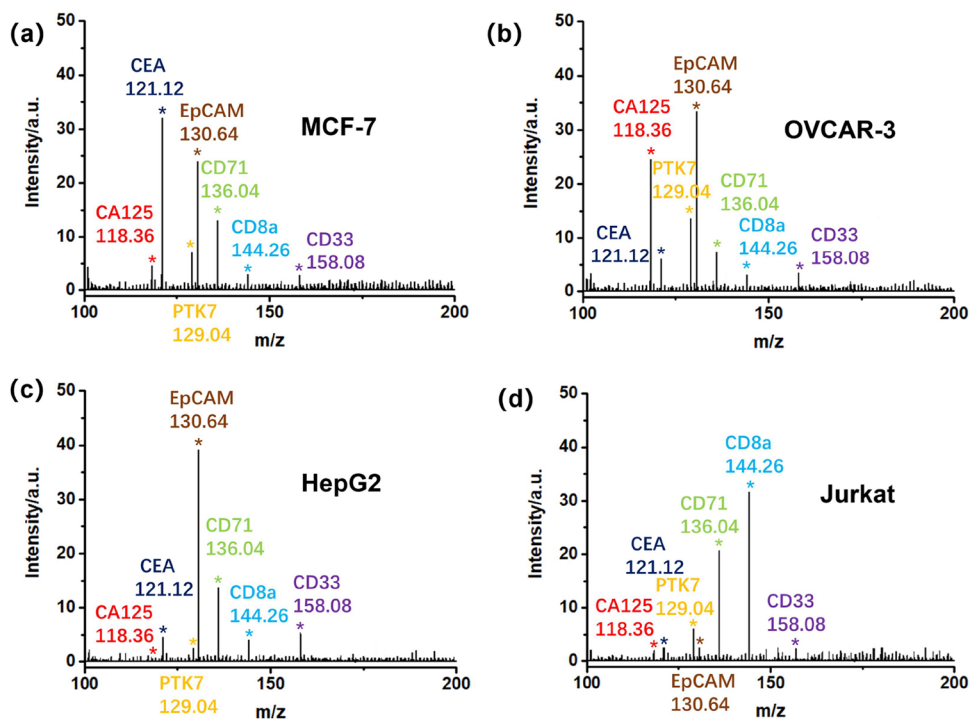


Figure 5. (a) Mass spectrum of mass tag indicative of MCF-7 cells, (b) of OVCAR-3 cells, (c) of HepG2 cells, and (d) and JURKAT cells.

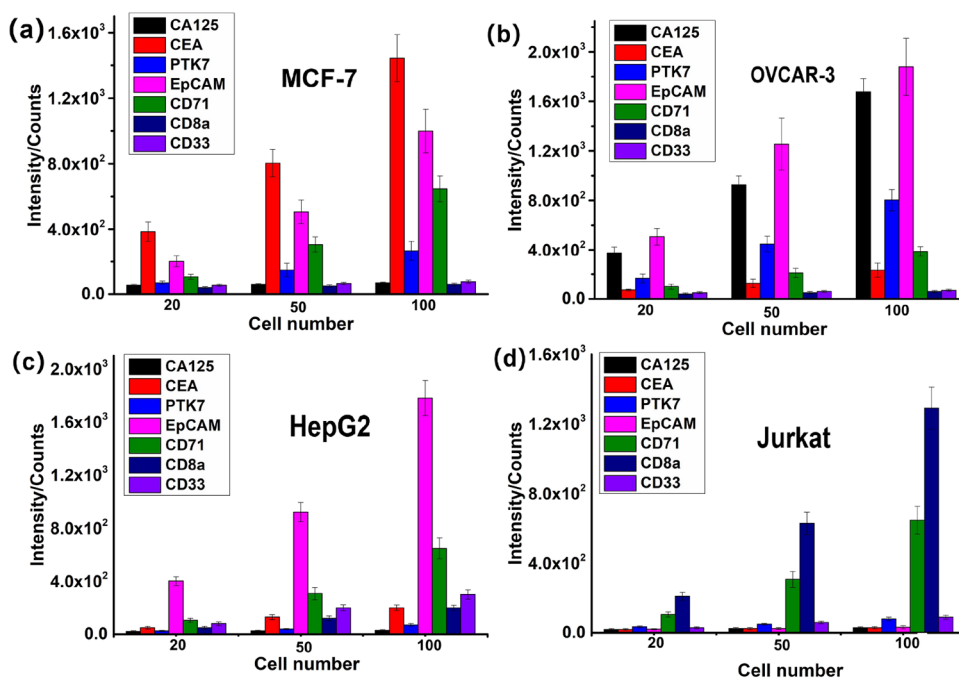


Figure 6. Multiplexed detection of CA125, CEA, EpCAM, PTK7, and CD71 on different numbers of MCF-7 cells (a), OVCAR-3 cells (b), HepG2 cells (c), and JURKAT cells (d) captured on the chips (RSD 9.8%, $n = 6$).

of these biomarkers were clearly discernible without any cross-interference.

As shown in Figure 5, the mass spectrum of mass tags indicative of five proteins on four different cancer cell types are present: (a) MCF-7 cells, (b) OVCAR-3 cells, (c) HepG2 cells, and (d) JURKAT cells. It can be seen in Figure 6 that CA125 is overexpressed in OVCAR-3 cells. CEA is overexpressed in MCF-7 cells. EpCAM is overexpressed in MCF-7, OVCAR-3, and HepG2 cells. PTK7 is overexpressed in

OVCAR-3 cells. CD71 is overexpressed in both MCF-7 and OVCAR-3 cells. CD8a is overexpressed in Jurkat cells. CD33 is not overexpressed in any of these cells. These results are consistent with previous reports.²³ In addition, we can distinguish different cell types by the differential expression of these proteins in different cells.

We further tested the specificity of the signal amplification assay, as shown in Figure S7a. The specificity experiment consists of three steps: first, incubate the aptamers (such as

CEA aptamers) intended for the specificity experiment with the cells. Second, incubate the cells with mAuNPs with aptamers, due to the binding of CEA to the aptamers in the first step; CEA is unable to bind to the mAuNPs. Finally, the signals are detected; there should be no signal indicating the presence of CEA protein. Figure S7 shows (b) the mass spectrum of the EpCAM specificity experiment in MCF-7 cells, (c) the mass spectrum of the EpCAM specificity experiment in OVCAR-3 cells, (d) the mass spectrum of the CEA specificity experiment in MCF-7 cells, and (e) the mass spectrum of the CEA specificity experiment in OVCAR-3 cells. (CA125, CEA, PTK7, EpCAM and CD71 were involved in the detection.) These results clearly illustrate the specificity of the strategy for the detection of protein biomarkers.

CONCLUSIONS

In this study, we have developed a signal amplification method with ultrahigh selectivity and sensitivity for detecting low-abundance protein biomarkers. The high sensitivity is attributed to DBDI's high ionization efficiency and ion transport ability for mass tag molecules. Consequently, the protein signals are extensively labeled and highly amplified (approximately 600 times). Additionally, this work successfully identified target biomarkers in complex biological samples (such as serum) and was able to analyze seven proteins on the cell surface in situ within 2 h. Organic small molecules are identified through MS with little signal overlap between parameters, and this advantage enables the possibility the simultaneous in situ detection of a much greater variety of proteins in the future. Our work not only provides a new strategy for mass spectrometry-based protein detection but also aids in differentiating between different types of cells. We believe that this method will pave the way for single-cell analysis, disease prediction, and tissue imaging using mass spectrometry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.4c06927>.

Chemicals and materials used; cell culture; description of the LA-DBDI-MS system; photograph of LA-DBDI-MS setup; TEM/EDS images of AuNPs; UV-vis spectra of modified AuNPs; mass spectra of MS tags; schematic diagram of MS tag detection; chromatography-based quantification of mass tags; protein-mass tag correspondence list; and specificity data for signal amplification assay (PDF)

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Notes

The authors declare no competing financial interest.

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